

ESTIMATING RUN-TO-RUN COVARIANCE ACROSS LANGUAGE MODEL BENCHMARKS

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ABSTRACT

Benchmark averages reduce training variability according to the covariance between benchmark scores, which per-benchmark standard deviations leave unspecified. We study this covariance with SNAP, a cross-half estimator that separates item-general run variation from item-sampling noise under conditional independence assumptions. We analyse 375 DataDecide runs across 125 configurations and ten benchmarks, scoring replicates at a common checkpoint step. Relative to independent benchmark deviations, the estimated standard deviation of the average increases by a factor of 1.244 for per-byte margins and 1.078 for accuracy, with nominal 95% recipe-cluster intervals of [1.143, 1.338] and [0.993, 1.157]. These inflation estimates depend on the chosen benchmark battery. Removing BoolQ raises the factors to 1.786 and 1.558, and shared passages remain an unresolved source of bias. Margin covariance concentrates within scoring formats, while gain simulations demonstrate sensitivity to score scale. A rank-one model has lower margin prediction error than full covariance, while full covariance has lower accuracy error, without uncertainty estimates for either difference. The planned recipe holdout was never implemented, so all empirical analyses are exploratory. These results support reporting replicate uncertainty and sensitivity to benchmark composition, while leaving a general independence approximation and a causal interpretation of margin covariance unresolved.

1 INTRODUCTION

A benchmark average can hide uncertainty from the training runs behind its component scores. For an equally weighted average of K benchmarks, independent run deviations give variance $K^{-2} \text{tr } \Sigma_E$, where Σ_E is their cross-benchmark covariance matrix. Covariance adds the off-diagonal terms, so a collection of marginal standard deviations doesn't determine the uncertainty of the average. Researchers need paired benchmark scores from replicate runs to estimate that contribution.

We estimate item-general covariance with two disjoint item halves for each benchmark. Within each configuration, we centre both half scores across replicate runs, then take their cross-half products. Under zero-mean item noise and conditional independence across halves, these products estimate run covariance, including its diagonal. The construction uses the logic of repeated measurements and variance components, rather than a new identity for correlated averages. We call its application to benchmark replicates SNAP, Seed Noise Across Phenotypes, and use phenotype to mean the measured score.

Our empirical study uses DataDecide (Magnusson et al., 2025), with 25 data recipes and five model sizes that each have three released replicates. The resulting 125 configurations supply 375 runs across ten benchmarks. We read released item scores, so the primary analysis requires no model inference. We compare per-byte log-likelihood margins with accuracy on the same items, because these scales respond differently to changes in confidence.

The distinction between a seed effect and run variation matters in this population. Auxiliary replicates use different training budgets below 1B parameters, and 92 configurations have mismatched final checkpoint steps. We compare the largest step shared by all three runs, but this rule doesn't align their training schedules. We therefore interpret the full estimate as covariance among the

released runs at the selected steps, and report an auxiliary-run contrast that removes the default-versus-auxiliary batch component.

The accuracy estimate is close to independence for the full battery, but that result changes with benchmark composition. BoolQ contributes 84% of the estimated accuracy variance trace, and questions from a shared passage can cross our item halves. Removing BoolQ increases the accuracy inflation factor from 1.078 to 1.558. This sensitivity prevents us from treating a small full-battery estimate as evidence that benchmark noise is independent in general.

Our contribution is an explicit estimand, a cross-half covariance construction with stated assumptions, and an empirical account of its limits on released language model runs. We compare score scales, examine batch and gain mechanisms, and test covariance models on held-out recipe folds. The initial analysis plan specified a separate screening set that we never formed. Its date has no independent timestamp, and none of the analyses supports a confirmatory interpretation.

2 ESTIMAND AND ESTIMATOR

2.1 RUNS AND SCORE SCALES

Let c index configurations, r index their R replicate runs, and j index benchmarks. A configuration fixes a data recipe, model size, and nominal hyperparameters. In DataDecide, the seed controls initialisation and data order together, while auxiliary training schedules introduce an additional source of run variation. For item i , define the per-byte margin and corresponding correctness indicator by

$$m_{crji} = \frac{\ell(\text{gold}_i)}{\text{bytes}(\text{gold}_i)} - \max_{v \neq \text{gold}_i} \frac{\ell(v_i)}{\text{bytes}(v_i)}, \quad a_{crji} = \mathbf{1}\{m_{crji} > 0\}. \quad (1)$$

Here ℓ is the summed answer log-likelihood, and v ranges over alternatives. This definition treats a zero margin as incorrect. We average items within each benchmark, with MMLU macro-averaged over its 57 subjects, and denote the resulting margin and accuracy scores by y^M and y^A .

For each half $H \in \{A, B\}$, our measurement model is

$$y_{crj}^H = \mu_{cj}^H + E_{crj} + \varepsilon_{crj}^H. \quad (2)$$

The half-specific mean allows the two item sets to differ in difficulty. The component E_{crj} is the run deviation shared by the two item populations, while ε_{crj}^H contains the remaining item-dependent variation. We assume independent, identically distributed run effects within a configuration when deriving unbiasedness, and write their covariance as $\Sigma_{E,c}$. The schedule differences in the actual population limit that interpretation.

2.2 CROSS-HALF COVARIANCE

We partition each benchmark’s items into disjoint halves with a frozen random seed, using the same assignment for its replicate runs. We centre each half score across the replicate runs, defining $d_{crj}^H = y_{crj}^H - R^{-1} \sum_s y_{csj}^H$. If the item errors have conditional mean zero and zero covariance between the two halves across the relevant runs and benchmarks, then

$$\mathbb{E}[d_{crj}^A d_{crk}^B] = (1 - 1/R) \Sigma_{E,c}(j, k). \quad (3)$$

The result includes diagonal entries, and doesn’t require Gaussian run effects. Disjoint item identifiers alone don’t establish the required independence, because passages, duplicate content, and other shared item effects can connect the halves.

We write the average centred score as $g_{cr}^H = K^{-1} \sum_j d_{crj}^H$ and compute

$$T_c = \frac{1}{R-1} \sum_r g_{cr}^A g_{cr}^B, \quad U_c = \frac{1}{K^2(R-1)} \sum_{j,r} d_{crj}^A d_{crj}^B. \quad (4)$$

The respective expectations of these two statistics are $K^{-2} \mathbf{1}^\top \Sigma_{E,c} \mathbf{1}$ and $K^{-2} \text{tr} \Sigma_{E,c}$. We sum each statistic across configurations before dividing, which gives

$$\hat{\Lambda} = \sqrt{\frac{\sum_c T_c}{\sum_c U_c}}, \quad \Lambda^2 = \frac{\mathbf{1}^\top \Sigma_E \mathbf{1}}{\text{tr} \Sigma_E} = 1 + (K-1) \bar{r}_E, \quad K_{\text{eff}} = \frac{K}{\Lambda^2}, \quad (5)$$

where $\Sigma_E = N^{-1} \sum_c \Sigma_{E,c}$ is the covariance matrix averaged across configurations. We express the off-diagonal contribution through the effective coefficient

$$\bar{r}_E = \frac{\sum_{j \neq k} \Sigma_E(j, k)}{(K-1) \text{tr} \Sigma_E}. \quad (6)$$

This effective covariance coefficient equals the mean off-diagonal correlation when the marginal variances are equal. With unequal variances, it differs from a normalised weighted mean correlation, as Appendix A explains. The effective count compares the battery with independent benchmarks at the same root-mean-square marginal deviation, rather than measuring its number of independent information sources.

The population inflation factor equals one when the aggregate off-diagonal covariance is zero. Unbiased estimates of T_c and U_c don't make their ratio or square root unbiased, and either pooled sum can be nonpositive in a finite sample. Calibration therefore checks the implemented estimator in specified populations, rather than proving an exact finite-sample null mean.

2.3 RESOLUTION AND INFERENCE

For an infinitesimal additive shift in Gaussian item margins, the ratio of accuracy to margin noise per unit squared sensitivity is

$$\frac{p(1-p)}{\phi(z_0)^2}, \quad z_0 = \Phi^{-1}(1-p). \quad (7)$$

At accuracy $p = 0.35$, this model gives a ratio of about 1.66. Its minimum is $\pi/2$, so the planned lower threshold of 1.4 couldn't falsify this model. This calculation describes the assumed location-shift model, and doesn't establish an empirical information advantage for arbitrary margin distributions. The aggregate reliability calculation uses $\rho_g = \Lambda^2/(\Lambda^2 + \bar{v})$, with $\bar{v}$ the item-noise contribution on that scale.

We report nominal 95% wild cluster bootstrap- t intervals over the 25 recipe clusters, using 4,999 Rademacher draws and a linearisation of the squared ratio. Appendix D states the implementation, including its fixed observed denominator during resampling. We also report a configuration percentile bootstrap, which resamples both sums but ignores recipe dependence. These procedures make different assumptions, and neither interval width alone identifies a design effect.

Reported permutation experiments give coverage of 0.925 for margins and 0.954 for accuracy under the permutation reference distribution. The margin interval falls below the null in 0.049 of replicates, above the nominal one-sided rate of 0.025. This result weakens a claim that its upper endpoint excludes the planned substantive threshold. It doesn't establish coverage at nonzero effects or under unmodelled passage dependence.

3 DATA AND ANALYSIS SCOPE

DataDecide supplies the ten benchmarks ARC-Challenge, ARC-Easy, BoolQ, CommonsenseQA, HellaSwag, MMLU, OpenBookQA, PIQA, Social IQa, and WinoGrande. We analyse 37,682 items per run across 25 recipes at 150M, 300M, 530M, 750M, and 1B parameters. Replicate seed labels are $\{2, 14, 15\}$ below 1B and $\{2, 4, 5\}$ at 1B. Smaller released models lack the required per-item replicate coverage.

We select the largest checkpoint step available for all three runs in each configuration. In 26 configurations, the largest final step exceeds the smallest by more than a factor of 1.5. The selected steps at 530M and 750M average 18% and 89% of the default run's final step, respectively. Conclusions from those bands concern truncated trajectories, rather than the released final models.

The author-reported plan assigned eight recipes to screening and seventeen to estimation, but no stage of the pipeline drew that partition. Analysis choices therefore used all 125 configurations, and the plan has neither a supplied original file nor an independently corroborated date. We retain its thresholds as historical targets, not as evidence of preregistration. An enumeration over the $\binom{25}{17} = 1,081,575$ possible estimation sets produces margin estimates from 1.037 to 1.317 and accuracy estimates from 0.963 to 1.191. None reaches the planned thresholds of 1.349 and 1.40, but

this retrospective calculation can't restore a holdout or account for choices made after viewing the data.

We retain the full ten-benchmark battery for the primary estimand, including benchmarks with negative estimated diagonal covariance. An unbiased moment estimate can be negative near zero, and deleting a benchmark after observing that sign changes both selection and the estimand. Standardised diagnostics use only defined positive-variance traits, so they carry this selection limitation.

We use PolyPythias (van der Wal et al., 2025) for an exploratory transfer check, scoring nine seeds at each of 70M, 160M, and 410M on 4,755 nested items. Its three configurations can't support a precise population comparison. A later five-size analysis adds 14M and 31M under a different scoring batch budget. We also examine released run-level scores from Heineman et al. (2025), whose single-configuration panels lack item-level outputs for the cross-half correction.

4 SENSITIVITY TO BATCH AND SCORE SCALE

The default and auxiliary runs come from different batches. We decompose within-configuration contrasts into a default-versus-auxiliary component and an auxiliary-versus-auxiliary component. The latter removes an offset shared by both auxiliary runs, while retaining one contrast per configuration. It doesn't establish matching software provenance or remove differences shared with the selected checkpoint schedules. A simulated batch offset provides a separate sensitivity check.

A positive multiplicative gain changes margin magnitudes while preserving their signs, so it leaves threshold accuracy unchanged by construction. We examine this mechanism with a gain covariate s_{cr} , the mean per-byte answer log-likelihood, and with simulations that vary the gain scale and margin distribution. We also fit a leave-one-benchmark-out competence proxy $x_{cr}^{(-j)}$ on the opposite item half, using

$$y_{crj} = \alpha_{cj} + \beta_j x_{cr}^{(-j)} + \gamma_j s_{cr} + u_{crj}. \quad (8)$$

The pooled fit uses one slope per benchmark over 250 run contrasts. These adjustments are descriptive, because the gain covariate shares item noise with the outcomes and the competence proxy uses unstable estimated variance normalisation. The implementation uses a standard-deviation floor of 10^{-6} when variance estimates are negative, making WinoGrande dominate the margin proxy by factors of about 800 to 38,000. A small change after this adjustment can't establish that general competence fails to explain the covariance.

5 RESULTS

5.1 FULL-BATTERY ESTIMATES AND COMPOSITION

Table 1 reports the full-battery inflation estimates on both score scales. The raw margin interval excludes one, while the accuracy interval includes it. The accuracy upper endpoint maps to $\bar{r}_E < 0.038$ and $K_{\text{eff}} > 7.47$ under the stated measurement model and fixed battery. Shared-passages bias prevents reading those values as unconditional bounds on item-general covariance.

Table 1. Inflation estimates across 125 configurations with nominal 95% recipe-cluster intervals. The auxiliary contrast compares the two auxiliary runs, and the adjustment includes the gain covariate alone. Aggregate standard deviations use margin units for margins and fractions for accuracy.

Score	Estimate	$\hat{\Lambda}$	Interval	K_{eff}	σ_{agg}
Margin	Raw	1.244	[1.143, 1.338]	6.46	0.00572
Margin	Gain adjusted	1.275	[1.111, 1.421]	6.15	0.00514
Margin	Auxiliary contrast	1.256	[1.090, 1.400]	6.34	0.00562
Accuracy	Raw	1.078	[0.993, 1.157]	8.60	0.00514
Accuracy	Gain adjusted	1.097	[0.980, 1.198]	8.31	0.00503
Accuracy	Auxiliary contrast	1.084	[0.939, 1.207]	8.51	0.00516

Per-benchmark reliability averages 0.652 on margins and 0.324 on accuracy. Seven accuracy benchmarks and three margin benchmarks have reliability below 0.50. WinoGrande has a negative estimated margin variance, while PIQA and WinoGrande have negative accuracy estimates. The accuracy correlation estimate also contains an entry of 1.296 and an eigenvalue of -0.55 , so it isn't a valid correlation matrix without further regularisation. The primary ratio uses covariance products directly, and doesn't require this normalised matrix to be positive semidefinite.

Removing BoolQ raises margin inflation to 1.786 with interval $[1.689, 1.877]$, and accuracy inflation to 1.558 with interval $[1.412, 1.687]$. Removing CommonsenseQA instead gives inflation estimates of 1.096 and 1.014. BoolQ contributes 68% of the margin covariance trace and 84% of the accuracy trace, while its covariance with other benchmarks is small. Its variance therefore has substantial weight in the denominator of Equation 5. Questions that share a BoolQ passage can enter different halves, and the analysis hasn't reconstructed passage groups from the public data. We can't separate that dependence from BoolQ's variance contribution with the reported outputs.

Recipe removal moves margin inflation between 1.207 and 1.256, while size removal moves it between 1.178 and 1.289. The corresponding accuracy ranges are $[1.065, 1.099]$ and $[1.047, 1.115]$. Benchmark-removal sensitivity is larger, but comparing that spread with a sampling standard error doesn't turn it into a probability distribution over benchmark collections. The design treats this battery as fixed, and supplies no sampling model for a wider benchmark population.

The defined-trait standardisation diagnostic gives inflation of 1.803 on nine margin traits and 1.707 on eight accuracy traits, without intervals. The latter maps to 2.75 effective benchmarks, compared with the full-battery accuracy count of 8.60. Both the weights and retained trait set change, so this contrast describes a different aggregate. Unstable denominators and selection on positive estimated variance limit its interpretation.

5.2 FORMAT STRUCTURE AND MECHANISM CHECKS

Figure 1 shows the estimated correlations and their eigenvalue spectra for both score scales.

Among the nine defined margin traits, mean estimated correlations are 0.624 within scoring formats and 0.007 across formats. Retaining only cross-format off-diagonal covariance gives a diagnostic inflation of 0.921 with interval $[0.801, 1.035]$, compared with 1.244 for the full matrix. The accuracy diagnostic is 0.965 with interval $[0.848, 1.063]$. Removing BoolQ gives a cross-format margin diagnostic of 1.006, compared with 1.786 without the restriction. Masking covariance entries needn't preserve positive semidefiniteness, so this diagnostic isn't the variance of a separately observed aggregate. Each benchmark appears in one format, which also prevents a causal separation of format from task content.

The auxiliary-only point estimates retain the full-battery inflation, but their intervals remain wide. Adjusting for gain increases margin inflation to 1.275 and accuracy inflation to 1.097. Joint adjustment gives 1.276 and 1.094, while competence adjustment alone gives 1.244 and 1.076. These results don't match the planned signature of margin inflation collapsing after gain adjustment, although the proxy limitations restrict any interpretation of that failure.

The default gain simulation at gain standard deviation 0.05 produces margin inflation of 1.148, about 61% of the observed excess above one. At the estimated gain standard deviation of 0.042, the default and distribution-matched simulators produce about 44% and 6% of that excess. These outcomes describe sensitivity to simulator choices, not a causal interval for the fraction explained by sharpness. Accuracy remains unchanged under positive scaling by definition, so its simulated invariance supplies no independent validation. The calibrated batch-offset simulation gives factors of 1.004 and 1.003, corresponding to about 1.5% of the observed margin excess.

5.3 CALIBRATION AND EXTERNAL CHECKS

The reported independent-noise simulation gives mean inflation of 1.0009 on margins and 1.0004 on accuracy. Independent per-trait run-label permutation gives 1.0010 and 0.9984. Across simulated effective covariance coefficients from zero to 0.50, the reported recovery slope is 0.998 with intercept 0.003. Fifty item re-splits have standard deviations of 0.002 and 0.008. These checks assess

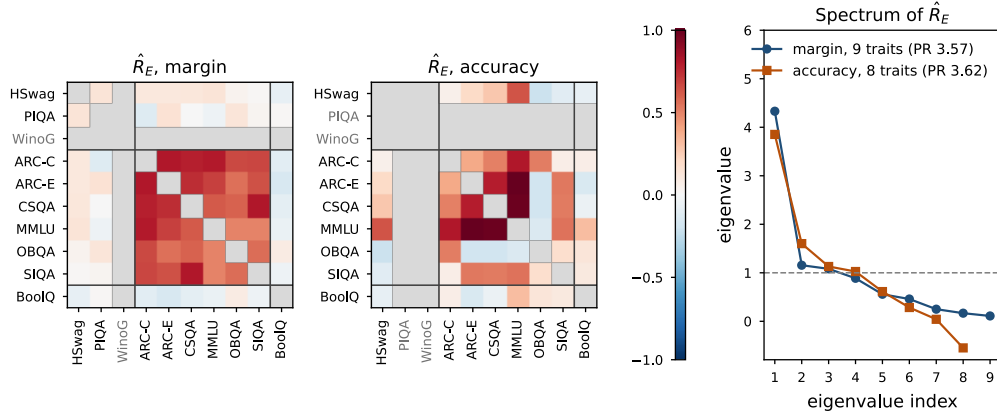


Figure 1. Cross-half correlation estimates across the 125 configurations, with undefined variances shown in grey. The accuracy estimate has a negative eigenvalue, so its spectrum is a diagnostic of the moment estimate rather than a valid population correlation spectrum.

selected reference distributions, and can’t rule out shared-passage dependence or training-schedule confounding.

The 500-permutation accuracy tail fraction is 0.028, while a later 1,000-permutation run gives 0.021. Both differ from the primary cluster interval, which contains one. Permutation exchangeability is stronger than a zero aggregate covariance condition, and the permutation and cluster procedures condition on different aspects of the design. We therefore report the disagreement without selecting the result with a smaller tail probability.

PolyPythias gives margin inflation of 1.406 on the three planned sizes, compared with 1.244 for DataDecide. The ratio of excesses above one is 1.66, but the ratio of inflation factors is about 1.13. The reported transfer gate used the first scale after the plan left the comparison ambiguous, so it can’t establish failure of a 20% aggregate-standard-deviation criterion. A five-size rescore gives margin inflation of 1.262, with interval $[0.927, 1.537]$, while its same-job three-size estimate is 1.414 with interval $[0.939, 1.781]$. Accuracy factors are 2.286 and 1.711 without intervals, which further limits a general independence recommendation.

The initialisation panel from Heineman et al. (2025) gives likelihood inflation of 1.705 and accuracy inflation of 0.989. Its data-order panel gives corresponding estimates of 1.625 and 0.868. These are single-configuration estimates on different likelihood and accuracy batteries, without cross-half correction or uncertainty intervals. They suggest a score-scale difference, but don’t independently confirm the DataDecide estimand.

5.4 PREDICTING AGGREGATE UNCERTAINTY

We compare five covariance models on five held-out recipe folds, giving each model the held-out marginal seed standard deviations. Table 2 therefore isolates the correlation component conditional on variance estimates that a practitioner would still need replicate runs to obtain. Negative variance estimates are clipped before the model branch, while the target retains the original covariance contributions, introducing a further evaluation convention.

The rank-one model has the lowest margin point error, while full covariance has the lowest accuracy point error. Without paired uncertainty estimates, these rankings don’t establish equivalence or a reliable gain from the extra parameters. The unrescaled phenotypic plug-in has the largest error on both scales. Rescaling its off-diagonals on training folds reduces that error, but requires replicate information and still has higher point error than the best model on either scale.

Table 2. Mean squared prediction error of log aggregate standard deviation across five recipe folds. No interval for a paired error difference accompanies these reported means. The rescaled plug-in is an exploratory addition to the original five-model comparison.

Model	Margin	Accuracy
Independence	0.0358	0.0143
Rank-one competence	0.0057	0.0140
Rank-one plus gain	0.0063	0.0260
Phenotypic correlation plug-in	0.1073	0.1682
Full covariance	0.0068	0.0105
Rescaled phenotypic plug-in	0.0131	0.0342

Under independent Gaussian run errors with a common aggregate standard deviation, a true score gap Δ reverses sign with probability

$$P_{\text{flip}}(\Delta) = \Phi\left(-\frac{\Delta}{\sqrt{2}\sigma_{\text{agg}}}\right). \quad (9)$$

Using the reported accuracy deviation of 0.00514 gives probabilities of about 0.246 for a half-point gap and 0.085 for a one-point gap. The independence approximation gives about 0.229 for a half-point gap. These are model-based calculations, rather than observed reversal rates, and paired training errors or unequal variances require a different denominator. The reported census places 455 of 1,500 observed same-size gaps below the corresponding 5% risk threshold, but noisy observed gaps don’t provide a lower bound on the fraction of small true gaps.

6 RELATED WORK

Bouthillier et al. (2021) derive variance inflation from correlated repeated trainings and estimate its scalar correlation in case studies. We apply covariance accounting across benchmarks within a replicate run. Reimers and Gurevych (2017) and Madaan et al. (2024) study seed-related evaluation variability, while Dodge et al. (2020) examine whether favourable initialisations transfer across tasks. Jordan (2024) separates example-specific training variance from distribution-level variation and compares accuracies across related evaluation sets. These precedents motivate a narrower contribution than a first measurement of any cross-dataset training dependence.

The cross-half construction follows repeated-measurement and variance-component ideas, including Spearman (1904), Cronbach (1951), and Searle et al. (1992). The quantitative-genetics distinction between phenotypic and residual covariance provides an analogy, but our model doesn’t identify separate genetic or shared-environment components. Cheverud (1988) motivates the phenotypic plug-in comparison, and its empirical failure here concerns this transfer of correlation structure between two sources of variation.

Heineman et al. (2025) study evaluation signal and noise, while Magnusson et al. (2025) and van der Wal et al. (2025) provide the replicate populations used here. Dehghani et al. (2021) show the consequences of benchmark composition, which our variance-weighted estimates also expose. Few-cluster inference follows the concerns studied by Cameron et al. (2008) and MacKinnon and Webb (2017), with calibration remaining specific to the implemented procedure and reference population. Appendix B discusses further related approaches and their different units of replication.

7 DISCUSSION AND LIMITATIONS

SNAP estimates cross-half covariance under a measurement model that separates item-general run effects from item-dependent noise. The empirical results show why both the model and benchmark weights need attention. Accuracy inflation is small for the full DataDecide battery, while a single benchmark removal produces a larger estimate. Margin covariance also changes with scoring format and gain simulation assumptions, so neither score scale supports an unqualified statement about independent benchmark noise.

The principal unresolved issues concern identification and inference. Shared BoolQ passages can bias the dominant diagonal term, common-step scoring leaves training-schedule differences, and the competence adjustment is dominated by variance flooring. The margin interval has imperfect coverage under the reported permutation reference, and prediction rankings lack paired uncertainty estimates. A held-out text loss would help assess score-scale mechanisms, but wouldn't by itself identify a causal competence component. The original screening split was never implemented, which leaves the empirical study exploratory throughout.

The reported adjacent-checkpoint experiment also limits a seed-only interpretation. Across nine PolyPythias 160M runs, moving 1,000 steps changes aggregate scores by 0.89 between-seed standard deviations on margins and 1.57 on accuracy. This narrow experiment doesn't quantify the corresponding effect at DataDecide's selected steps, but demonstrates that checkpoint choice can matter at the scale of the variation under study.

For evaluation practice, we recommend paired replicate scores across the chosen battery, an aggregate seed standard deviation, and a benchmark-removal sensitivity check. A small estimated covariance contribution doesn't make a small score gap stable across runs. The present evidence supports a conditional description of this battery, while additional passage-aware splits and reproducible paired model comparisons are needed before recommending an independence approximation more broadly.

AI USE STATEMENT

We used generative AI tools to propose and refine hypotheses, critique the estimator and simulation design, assist with analysis code, revise manuscript text, and survey literature. AI assistance also contributed to the technical and editorial review of this version. The study includes synthetic calibration and sensitivity simulations, whose role differs from the released model outputs used for the empirical analysis. The authors remain responsible for verifying AI-assisted claims, code, citations, and the final submission, including an accurate disclosure of AI involvement in generating simulation data.

ETHICS STATEMENT

This study analyses released model outputs and introduces no new participant recruitment or data collection. Its intended use is to improve uncertainty reporting in model evaluation. An independence approximation used outside the studied battery could understate uncertainty, so the recommendations remain conditional on the measured population and score definition. Use of derived artifacts must retain the applicable source attribution and licensing requirements.

REPRODUCIBILITY STATEMENT

Section 2 and Appendix A define the estimator and its assumptions, while Appendix D specifies the interval calculation and data reduction. Appendix B records the populations and planned checks, and Appendix C reports later sensitivity analyses alongside the primary results. The primary calculation uses 4,999 bootstrap draws, 2,000 calibration replicates, 500 permutation replicates, 50 item re-splits, and master seed 20260101. Reproduction requires the per-item reductions, item ordering and half assignments, selected checkpoint identifiers, and the corresponding analysis scripts. The original analysis plan lacks an independently corroborated timestamp, and its proposed screening split wasn't implemented. The additional simulation, transport, and checkpoint analyses require their own execution records, rather than the primary reduction alone.

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A DERIVATIONS AND INTERPRETATION

A.1 WHAT SNAP ESTIMATES

SNAP abbreviates Seed Noise Across Phenotypes, with a phenotype denoting a benchmark score. Its target is the covariance of run deviations that generalise across item halves. DataDecide replicates vary initialisation and data order together, and auxiliary runs below 1B use shortened training schedules. Common-step scoring aligns the observation step, while schedule differences remain within the full-set target. We therefore use run covariance for the empirical estimate, reserving a seed-only interpretation for designs with matching training conditions.

The distinction between item-general variation and fixed-battery variation determines how a practitioner can use the estimate. A run can perform differently on particular items without changing its expected score over an item population. Under the measurement model, that variation enters the item error and doesn't contribute to the cross-half covariance target. Repeated evaluation of a fixed item bank can retain this component, so its run variance needn't equal the item-general variance estimated here.

The analogy with quantitative genetics distinguishes covariance between configurations from residual covariance within them. It doesn't identify separate additive-genetic and shared-environment components in a model-training experiment. Between-configuration correlation R_P reflects differences in recipe and model size, while the within-configuration covariance Σ_E concerns replicate deviations. The plug-in comparison asks whether the first correlation structure predicts the second after supplying marginal run variances.

A.2 NOTATION

Table 3. Notation for the cross-half construction and its empirical diagnostics. The coefficient $\bar{r}_E$ uses covariance normalisation, which differs from mean correlation under unequal marginal variances.

Symbol	Meaning
c, r, j, i	Configuration, replicate run, benchmark, and item indices
N, R, K	Configuration, replicate, and benchmark counts, with values 125, 3, and 10
$\Sigma_{E,c}, \Sigma_E$	Configuration covariance and its average across the design
R_E, R_P	Run-effect correlation and between-configuration score correlation
y^M, y^A	Mean per-byte margin and accuracy on the specified item set
$d_{cr,j}^H, g_{cr}^H$	Centred benchmark score and its benchmark average on half H
T_c, U_c	Cross-half aggregate covariance and its diagonal-only contribution
$\Lambda, \bar{r}_E$	Standard-deviation inflation and effective covariance coefficient
K_{eff}	Independent-benchmark equivalent at the same marginal RMS deviation
$\sigma_{\text{agg}}, \sigma_{\text{indep}}$	Aggregate deviation with full covariance and under independence
$\rho_g, \bar{v}$	Aggregate reliability and its scaled item-noise contribution
$s_{cr}, x_{cr}^{(-j)}$	Gain covariate and opposite-half competence proxy

A.3 CROSS-HALF IDENTITY AND ITS ASSUMPTIONS

Fix a configuration and suppress its index for clarity. Write the half-score model as $y_{rj}^H = \mu_j^H + E_{rj} + \varepsilon_{rj}^H$, so centring across runs gives

$$d_{rj}^H = (E_{rj} - \bar{E}_{\cdot j}) + (\varepsilon_{rj}^H - \bar{\varepsilon}_{\cdot j}^H). \quad (10)$$

Assume that run-effect vectors are independent draws with covariance Σ_E . Conditional on those effects, assume zero-mean item errors and zero covariance between ε_{rj}^A and ε_{sk}^B for the run and

benchmark pairs in the calculation. Expanding $d_{rj}^A d_{rk}^B$ then removes its error cross terms in expectation, leaving

$$\mathbb{E}[d_{rj}^A d_{rk}^B] = \mathbb{E}[(E_{rj} - \bar{E}_{\cdot j})(E_{rk} - \bar{E}_{\cdot k})] \quad (11)$$

$$= (1 - 1/R) \Sigma_E(j, k). \quad (12)$$

Summing across runs and dividing by $R - 1$ gives the unbiased covariance moment. We symmetrise the two half orderings for the matrix estimate,

$$\hat{\Sigma}_E(j, k) = \frac{1}{2N(R - 1)} \sum_{c,r} (d_{crj}^A d_{crk}^B + d_{crk}^A d_{crj}^B). \quad (13)$$

Symmetry follows from this construction, while positive semidefiniteness doesn't follow for a finite sample. A negative diagonal moment near zero therefore doesn't by itself falsify the identity, although it prevents ordinary correlation normalisation for that trait.

The independence assumption concerns item effects, rather than whether two identifiers differ. Shared source passages, duplicate questions, and overlapping content can create cross-half covariance. A fixed finite item bank split without replacement also requires a model for its sampling dependence. The available item-half assignment alone can't verify these conditions.

BoolQ is a specific unresolved case in this analysis. Questions associated with one passage can enter both halves, and a run-level response to that passage can correlate their errors. The original analysis didn't join public passage text to the retained item identifiers. Its covariance estimate can therefore include a passage-dependent diagonal contribution that the item-general target excludes. Removing BoolQ changes both this exposure and the benchmark composition, so the removal sweep can't identify the size of the leakage alone.

For a diagonal additional covariance D with nonnegative trace, the inflation ratio becomes

$$\Lambda_D^2 = \frac{\mathbf{1}^\top \Sigma_E \mathbf{1} + \text{tr } D}{\text{tr } \Sigma_E + \text{tr } D}. \quad (14)$$

This ratio lies between Λ^2 and one when the denominators are positive. It explains attenuation toward one under that restricted model, but doesn't yield a general upper or lower bound when shared-item effects also contribute off-diagonal covariance. The same restriction is needed before using the item-general factor to bound fixed-battery inflation.

A.4 COVARIANCE COEFFICIENT AND EFFECTIVE COUNT

Let $\sigma_j^2 = \Sigma_E(j, j)$ and $r_{jk} = \Sigma_E(j, k) / (\sigma_j \sigma_k)$ whenever both variances are positive. For this finite battery, the variance ratio therefore satisfies

$$\Lambda^2 = 1 + \frac{\sum_{j \neq k} r_{jk} \sigma_j \sigma_k}{\sum_j \sigma_j^2} = 1 + (K - 1) \bar{r}_E. \quad (15)$$

The normalised weighted mean correlation instead divides the numerator by $\sum_{j \neq k} \sigma_j \sigma_k$. For a positive denominator, the two coefficients are related by

$$\bar{r}_E = a_\sigma \bar{r}_{\text{weighted}}, \quad a_\sigma = \frac{\sum_{j \neq k} \sigma_j \sigma_k}{(K - 1) \sum_j \sigma_j^2}. \quad (16)$$

For nonnegative standard deviations, a_σ lies between zero and one. With two perfectly correlated traits having standard deviations one and two, the effective coefficient is 0.8 even though the weighted mean correlation is one. Equal marginal variances make the coefficient and the mean correlation coincide.

With average marginal variance $\bar{\sigma}_E^2 = K^{-1} \text{tr } \Sigma_E$, the variance of the benchmark average satisfies

$$\sigma_{\text{agg}}^2 = \bar{\sigma}_E^2 \left[\frac{1}{K} + \frac{K - 1}{K} \bar{r}_E \right]. \quad (17)$$

A nonzero asymptotic floor follows only for a sequence of batteries whose average variance and effective coefficient converge to suitable limits. Bounded variances alone don't imply that convergence, and the present ten-benchmark study doesn't estimate such a sequence. We use the finite-battery identity for the reported effective counts.

For the raw, gain-adjusted, and auxiliary-contrast rows, the reported effective margin coefficients are 0.0608, 0.0695, and 0.0641. The corresponding accuracy coefficients are 0.0181, 0.0225, and 0.0194, retaining the covariance normalisation of Equation 6.

For positive aggregate variance, $K_{\text{eff}} = K/\Lambda^2$ compares the measured average with an independent average having the same marginal RMS deviation. Negative aggregate off-diagonal covariance can give an effective count above the benchmark count, $K_{\text{eff}} > K$. A single positive-variance benchmark has $\Lambda = 1$, although its off-diagonal coefficient is undefined because $K - 1 = 0$. This arithmetic count also differs from a spectral participation ratio.

A.5 POOLING AND FINITE-SAMPLE BIAS

We pool T_c and U_c across configurations before taking their ratio. Each individual denominator uses two replicate degrees of freedom and can approach zero or become negative. Averaging individual ratios would give those unstable denominators disproportionate influence. Pooling reduces that problem but doesn't remove the effect of configurations with large covariance contributions.

The reported recipe influence participation ratios are 2.88 for margins and 8.68 for accuracy. One recipe contributes 0.543 of the squared margin influence, while removing that recipe moves the estimate by at most 0.037. These quantities describe influence concentration, and aren't literal counts of independent clusters for inference. The reported permutation coverage of 0.925 on margins is a reason to qualify the nominal interval, rather than evidence that a bootstrap automatically resolves concentration.

The covariance moments are unbiased under their assumptions, while the pooled ratio has a random denominator. Its square root introduces another nonlinear transformation, so neither transformation inherits exact unbiasedness. A finite implementation must also specify what happens when a pooled sum is nonpositive. The reported data give finite full-battery estimates, but that observation doesn't establish finite estimates for arbitrary input populations.

A.6 PERMUTATION REFERENCE

For each configuration and trait, apply a uniform permutation π_j of run labels to both halves together. The diagonal products then sum to the same value, leaving U_c unchanged. Independent permutations across traits give zero expected off-diagonal products because each centred deviation sums to zero. Conditional on a positive fixed pooled denominator, this implies

$$\mathbb{E}_\pi \left[\frac{\sum_c T_c^\pi}{\sum_c U_c} \right] = 1. \quad (18)$$

This expectation for the squared ratio doesn't imply that $\mathbb{E}_\pi[\hat{\Lambda}^\pi] = 1$. If all permuted ratios are non-negative, concavity of the square root gives an expectation no larger than one, with strict inequality for a nonconstant ratio. For example, equally likely ratios of 0.5 and 1.5 average to one, while their square roots average to about 0.966.

A common permutation across all traits preserves their pairings and can't remove cross-trait association. Independent trait permutations avoid that problem, and the identity belongs in the uniform permutation sample space. Its chance of reproducing the observation doesn't invalidate a permutation procedure. The construction preserves marginal score values and cluster membership, but can alter dependence of run alignments across configurations, so validity as a test requires an exchangeability assumption beyond zero mean covariance.

A.7 PLANNING CALCULATIONS AND INFORMATION RATIO

The author-reported power calculation assumes approximately Gaussian half-score deviations. With $d^A = e + a$, $d^B = e + b$, independent errors a, b , and $\text{Var}(a) = \text{Var}(b) = k \text{Var}(e)$, the product variance is

$$\text{Var}(d^A d^B) = \sigma^4[(1+k)^2 + 1], \quad k = 2(1 - \rho_g)/\rho_g. \quad (19)$$

The planning values used two replicate degrees of freedom per configuration, numerator-denominator correlation 0.7, and design effect 1.60. Appendix B preserves the resulting values

for the intended 85-configuration analysis, which differs from the realised 125-configuration study. We don't infer realised power from the observed effect.

For the information calculation, let item margins follow a Gaussian location model with standard deviation s_m . An infinitesimal shift δ changes accuracy by $f(0)\delta + o(\delta)$, while it changes the mean margin by δ . Under independent item sampling, the item-noise variances are $p(1-p)/n$ and s_m^2/n , and the Gaussian model gives $f(0) = \phi(z_0)/s_m$. Dividing noise by squared local sensitivity yields Equation 7. The scale cancellation and minimum $\pi/2$ belong to that model, rather than arbitrary empirical margin distributions.

The competence and gain adjustments have additional measurement limitations. The opposite-half competence proxy uses estimated seed standard deviations, including a 10^{-6} floor where the margin variance estimate for WinoGrande is -3.15×10^{-7} . The gain covariate uses both halves, so its item noise can correlate with the outcome errors. The reported zero-mediation simulation treats this shared item component differently from the observed regression. Its margin reference moves from 1.24502 to 1.0910, with standard deviation 0.096, while the fully adjusted observed value is 1.276. The accuracy reference begins at 1.11831 rather than the observed 1.07837, and ends at 1.1045 against an adjusted 1.094. These mismatches prevent a calibrated causal interpretation of the residualisation comparisons.

B DESIGN DETAILS

B.1 COMMON STEP RULE

For each of the 125 configurations, we selected the largest released checkpoint step available for all three runs. The analysis records that step in each reduced filename, although the supplement described in the analysis record omits the fetch stage's complete configuration table. In 92 configurations, the three runs have different maximum steps. DataDecide stops the two auxiliary runs below 1B at 25 percent of the 1B compute budget and trains three complete runs only at 1B (Magnusson et al., 2025). The per-instance release contains auxiliary runs from 150M upward, so the five replicated sizes used here begin at 150M rather than 4M. This availability rule explains the difference from DataDecide's fourteen-size ladder.

In 26 configurations, all at 530M and 750M, the largest maximum step exceeds the smallest by more than a factor of 1.5. The 18 and 89 percent truncation figures reported for those sizes describe aggregates over 25 configurations per band, rather than individual configurations. We assess the resulting sensitivity through the auxiliary-run contrast and the estimates by size in Table 8.

B.2 RELIABILITY PHENOTYPES

The item halves contain 18,840 and 18,842 items across the battery. The fetch stage creates a balanced split within each trait and records the split for reuse across runs. The planning document proposed estimating reliability on eight screening recipes before inspecting seventeen estimation recipes. We didn't implement that separation, and Table 4 instead uses all 25 recipes. The reliability checks and the decisions based on them therefore remain exploratory.

Two accuracy traits have negative reliability estimates, and two others fall below 0.10. The pipeline summary reports an information-ratio median of 1.61, a minimum of 1.571, and a maximum of 1.732, without releasing the individual ratios. The battery-level ρ_g estimate is 0.73, compared with the planning value of 0.49 at $\Lambda^2 = 1.819$. The reported matrices contain the mean variance used to reconcile those quantities.

B.3 POWER CALCULATIONS

Table 5 reproduces the planning arithmetic for 85 configurations and seventeen recipe clusters. The completed analysis instead uses 125 configurations and 25 clusters. On the Λ scale, the reported linearised standard errors are 0.045 for margins and 0.038 for accuracy, compared with planning values of 0.0935 and 0.1232. The corresponding variance-ratio standard errors, 0.112 and 0.081, belong to the Λ^2 scale and aren't comparable with the planning column.

Table 4. Cross-half reliability of run deviations across all 25 recipes. Reliability equals $\hat{\sigma}_E^2/(\hat{\sigma}_E^2 + \hat{\sigma}_{\text{noise}}^2)$ and can be negative when the estimated run component is negative. MC denotes a multiple-choice scoring format in this table.

Trait	Format	Accuracy	Margin
ARC-Easy	MC	0.602	0.927
ARC-Challenge	MC	0.203	0.584
BoolQ	yes/no	0.970	0.993
CommonsenseQA	MC	0.615	0.930
HellaSwag	cloze	0.450	0.757
MMLU	MC	0.022	0.860
OpenBookQA	MC	0.091	0.392
PIQA	cloze	-0.028	0.273
Social IQa	MC	0.416	0.888
WinoGrande	cloze	-0.102	-0.086
Battery mean		0.324	0.652

The realised standard errors are smaller by factors of 2.1 and 3.3. Increasing the configuration count alone would predict a factor of approximately 1.21 under otherwise fixed assumptions. Interpolation of the N1 simulation grid gives dispersion at the planned $\bar{r}_E = 0.0911$ that is 1.04 times the dispersion at the observed margin value of 0.061, and 1.13 times the dispersion at the accuracy value of 0.0181. Differences in the design effect and reliability also affect this comparison, but we haven’t separated their contributions. We retain the original planning calculations rather than calculating power from the estimated effect after observing the results.

Table 5. Planning calculations at $\Lambda^2 = 1.819$ and $N = 85$ configurations. The calculations use $t(16)$, a design effect of $1 + 4\rho_{\text{ICC}} = 1.60$ at $\rho_{\text{ICC}} = 0.15$, and $\rho_g = 0.61$ for margins or 0.49 for accuracy.

Phenotype	Aggregation	Planned SE(Λ)	MDE(Λ)	Power
Margin	Cross-half	0.0935	1.300	0.89
Margin	Battery mean	0.0845	n/a	0.94
Accuracy	Cross-half	0.1232	1.412	0.68
Accuracy	Battery mean	0.1076	n/a	0.79
Accuracy	Cross-half, $\bar{r}_E = 0.115$	0.1232	n/a	0.82

B.4 DECISION CHECKS AND DEVIATIONS

The planning document lists nine checks with proposed responses. Its claimed date is September 5, 2026, but we haven’t established that date through an independent timestamp. We consequently describe the document as a reported plan and treat its checks as exploratory assessments.

A batch offset can contribute to the same cross-trait products as run-level variation. The auxiliary-run contrast removes the shared default-versus-auxiliary contrast from that estimate, while the estimate using all three runs remains exposed. Check G8 passes on point estimates, although the intervals cross the decision boundaries. We haven’t corrected the full-set estimate for this exposure.

Under the default gain simulator, $\sigma_g = 0.05$ produces approximately 61 percent of the observed margin excess. An empirically matched simulator produces approximately eight percent at the same gain scale and six percent at the estimated scale of 0.042. These results describe sensitivity to simulator assumptions and aren’t causal bounds on the contribution of gain. The accuracy result offers no independent protection against this mechanism because multiplying a margin by a positive factor leaves its sign unchanged. Partialling the gain covariate increases Λ_M , and adding gain to the rank-one predictor worsens held-out point error relative to the scalar predictor alone. These observations don’t identify a unique causal explanation.

Table 6. Planning checks, proposed responses, and observed outcomes in the source analysis. The original assessment labels four checks as failures, while G2 uses a lower threshold that can’t discriminate under its model.

Check	Condition and timing	Proposed response	Reported outcome
G0	Outputs parse at the common step, day 1	Drop unavailable cells, report reduced N if fewer than 70 estimation cells remain	Pass, all 125 cells
G1	Accuracy power at $\bar{r}_E = 0.115$ at least 0.80, day 2	Make accuracy secondary and report margin in the abstract	Pass, power 1.00 at $\Lambda_{\text{alt}} = 1.427$, $\rho_g = 0.73$
G2	Median information ratio at least 1.4, day 2	Remove comparative information claim	Ratio 1.61, threshold below the ratio’s floor
G3	Symmetric cross-half covariance and positive diagonal, day 3	Split by source document, then report leakage if asymmetry persists	Fail, one negative margin diagonal and two accuracy diagonals
G4	$ \hat{\Lambda} - 1 \leq 0.02$ under N1 and N2, day 4	Subtract calibrated offset and report both estimates	Pass, offset at most 0.002
G5	N5 accounts for less than 50 percent of excess, day 5	Lead with sharpness and report accuracy alone	Fail, default simulator share 0.61
G6	PolyPythias transport error within 20 percent, day 6	Report transport failure and its ratio	Fail upward under the excess-ratio reading
G7	Held-out loss available in metadata, day 1	Use leave-one-trait-out competence covariate	Fail, no required field
G8	Auxiliary-run $\Lambda \geq 1 + 0.5(\Lambda_{\text{full}} - 1)$, day 5	Lead with auxiliary-run estimate if condition fails	Pass on both point estimates, margin 1.256 versus 1.122, intervals cross their boundaries

The analysis didn’t execute the proposed responses for G3 and G5. G3 combines asymmetry with negative variance estimates, although a negative moment estimate doesn’t establish leakage. The available records also lack a usable source-document identifier for the proposed alternative split. We retain the margin analysis after the G5 failure with the simulator sensitivity disclosed, which departs from the proposed response. The source gate table contains seven rows, with transport and the auxiliary-run contrast supplying G6 and G8 separately.

B.5 REPORTED PLANNING PREDICTIONS

The planning document specifies eleven predictions and their rejection conditions. Table 7 records the outcomes without treating the document as verified preregistration.

- Both $\Lambda_M > 1.349$ and $\Lambda_A > 1.40$, with lower confidence limits above one, constitute the first planned prediction.
- The accuracy estimate retains at least 80 percent of the margin estimate, expressed as $\Lambda_A \geq 0.8\Lambda_M$.
- Accuracy retains more than 70 percent of its excess after gain adjustment, and N5 generates less than 30 percent of the margin excess.
- At measured token-budget ratios, N6 generates less than 50 percent of the full-set excess.
- The independence calculation underpredicts aggregate seed standard deviation by Λ in at least four of five size bands.
- DataDecide’s R_E predicts PolyPythias aggregate seed standard deviation within 20 percent, while independence misses by more than 20 percent.
- At least one seed mechanism in the external panel gives $\Lambda > 1$ and a bootstrap lower limit above one.
- At least eight traits have an information ratio between 1.4 and 3.0 in the planned assessment.

9. The cross-format diagnostic exceeds 1.2, with its lower confidence limit also exceeding one.
10. The Frobenius distance $\|R_E - R_P\|_F$ exceeds 0.5 in the comparison between the two correlation matrices.
11. More than ten percent of same-size configuration pairs have $P_{\text{flip}} > 0.05$ at the measured aggregate noise scale.

B.6 EXCLUDED POPULATIONS

MultiBERTs provides 25 seeds for one configuration (Sellam et al., 2022), whereas the 100 BERT runs of McCoy et al. (2020) vary fine-tuning from one pretrained checkpoint. Pythia’s size ladder lacks seed replicates (Biderman et al., 2023), and the OLMo-2 checkpoints considered here lack replicates at matched steps (Team OLMo et al., 2025). The signal-and-noise panel lacks per-item outputs, which limits its use to the external check. These restrictions concern the replication required for this estimator.

B.7 RELATION TO PRIOR MEASUREMENT WORK

Reimers and Gurevych (2017) compare score distributions across seeds within datasets, while Madaan et al. (2024) report benchmark-level seed variance. Hagmann et al. (2023) model non-determinism with scalar variance components and their interactions with data properties. Our target adds covariance between benchmark-specific run deviations to that measurement problem.

Jordan (2024) attributes much test-set variance to per-example variation across seeds, and its Appendix C compares run accuracy across five ImageNet evaluation sets with R^2 at most 0.14. Those comparisons concern shifted versions of one benchmark. Bui et al. (2025) vary fine-tuning seeds, and Hochlehnert et al. (2025) vary decoding seeds, which address different variance components from the pretraining-run variation examined here. Dodge et al. (2020, Section 4.3) ask whether a seed performs well across datasets and compare four GLUE task rankings without estimating a covariance matrix.

Zhou et al. (2020, Figure 3) report a 15×15 Spearman matrix across datasets. Their standard block has correlations at least 0.95, while standard-to-analysis entries include 0.97 for STR-L and 0.30 for STR-O. These correlations compare checkpoints within runs and average across seeds, so they combine changes in checkpoint performance rather than isolating within-configuration seed deviations. Dufour et al. (2026) separate training and sampling variation in FID, while Miller (2024) estimates item-level uncertainty for one model. Zhao et al. (2026) report the three-seed standard deviation of a twelve-task average and combine twelve per-task Welch tests in their Appendix E.4 without estimating cross-task covariance.

Haupt et al. (2026) use the PolyPythias 410M panel to estimate per-item variation and prioritise items by welfare weight and improbability. Their per-item variance estimates and approximate item-independence assumption address a different quantity from the off-diagonal covariance of benchmark run effects. This distinction concerns the estimand, even where the underlying model panel overlaps.

Cheverud (2001), Nyholt (2004), and Li and Ji (2005) estimate effective numbers of independent tests from observed correlation matrices. Cheverud uses eigenvalue variance, Nyholt uses spectral decomposition of linkage disequilibrium, and Li and Ji revise the effective-count approximation. Their application adjusts significance thresholds, whereas ours converts an estimated noise covariance into an aggregate standard deviation and an effective benchmark count. Searle et al. (1992) explain negative variance-component estimates from moment methods when the true component lies near zero.

Schaeffer et al. (2023) show how discontinuous metrics can change the appearance of performance trends. Accuracy thresholds our margin at zero, so we report both scales while retaining their separate noise properties. Low estimated seed signal relative to item noise provides a measurement limitation for accuracy in this study. Dehghani et al. (2021) show that benchmark selection can reorder methods, which concerns the composition sensitivity examined here.

Demšar (2006) and Dror et al. (2017) develop inference across datasets, while Card et al. (2020) examine statistical power and Gorman and Bedrick (2019) examine dependence on a single split. Burnell et al. (2023) advocate per-instance reporting alongside aggregate scores. Henderson et al. (2018) and Agarwal et al. (2021) address seed sensitivity and uncertainty in reinforcement learning. The authors resample runs within tasks, with an additional task-resampling variant in their Appendix A.5, while our application requires preservation of paired run effects across benchmarks.

Hardy et al. (2026) apply generalizability theory to 4,000 released models, which places replication across models rather than training seeds. Kohli (2026) estimates roughly two independent votes among nine frontier judges, and Messing (2026) reports naive standard errors that are 40 to 60 percent too small after accounting for judge and prompt variation. These examples involve related dependence calculations on different measurement axes.

The cross-half identity builds on the repeated-measurement approach to attenuation described by Spearman (1904). Under independent errors between halves, paired measurements remove the item-noise contribution from covariance estimates, including diagonal entries. Cronbach (1951) relates split-half reliability to random halves of an item population, which motivates the reliability table and the 50 re-splits in N3. Those re-splits change the margin estimate by a reported standard deviation of 0.002.

The Brown (1910) and Spearman (1910) test-length formula contains $1 + (k - 1)\rho$, the same algebraic form used for Λ^2 . Its correlation concerns parallel items, whereas ours concerns benchmark run effects. Substituting the reliability means in Table 4 at $k = 10$ would produce 6.87 and 3.91, which aren’t estimates of our variance inflation.

Campbell and Fiske (1959) compare traits and measurement methods in a crossed matrix. The within-format and cross-format summaries here resemble heterotrait blocks, but no benchmark appears in multiple formats. This design therefore can’t separate benchmark content from scoring format or establish format as the cause of the association. Brennan (2001) provides the variance-component framework for these distinctions.

Hofmann et al. (2025) fit benchmark-specific item-response models to an overlapping evaluation panel, with seven authors shared with Heineman et al. (2025). Kipnis et al. (2024) estimate a factor across six benchmarks that correlates 0.94 with total score. Those ability structures describe performance differences rather than the within-configuration noise covariance estimated here. Hedges et al. (2010) estimate uncertainty for dependent effect sizes without requiring their complete covariance matrix, while our application estimates the covariance needed for an effective-count point estimate.

C SUPPLEMENTARY RESULTS

Covariance diagnostics The largest discrepancy between opposite cross-half covariance entries is 2.44 bootstrap standard errors. This maximum across multiple pairs doesn’t by itself establish leakage. The available item records provide 57 MMLU subject groups but only one group for each other trait, so they don’t support the proposed source-document split.

Among the eight accuracy traits with positive estimated seed variance, estimated correlations range from -0.216 to 1.296 , with the maximum occurring for ARC-Easy and MMLU. The symmetrised matrix has a negative eigenvalue of -0.55 and a leading eigenvalue of 3.85 . These estimates don’t form a valid correlation matrix, a possible finite-sample outcome of noise-corrected moment estimation. We retain the unprojected matrix because a positive-semidefinite projection would change the reported values without an established correction target. The primary ratio uses cross-half sums directly and doesn’t require this correlation-matrix transformation. The nine-trait margin matrix has correlations from -0.157 to 0.802 and nine positive eigenvalues. Its leading eigenvalue is 4.33 , with a spectral participation ratio of 3.57 . The accuracy participation ratio is 3.12 before clipping its negative eigenvalue and 3.62 after clipping. Because that accuracy matrix is indefinite, neither figure provides a validated count of independent population directions.

The original scorecard counted five failures, four passes, and two predictions that couldn’t be tested as written. Predictions 9 and 11 came from a separate analysis of the per-run reductions, rather than the original pipeline. Prediction 7 lacks its required lower confidence limit, and prediction 6

Table 7. Exploratory scorecard for the eleven reported planning predictions. Several outcomes depend on interpretations absent from the planning document, and the table records those limitations.

No.	Criterion	Measured result	Assessment
1	Margin above 1.349, accuracy above 1.40, lower limits above 1	1.244 and 1.078	Neither point threshold reached
2	Accuracy-to-margin ratio at least 0.8	0.867	Point criterion met
3	Retained excess above 0.70 and N5 share below 0.30	1.23 and 0.61	Default N5 criterion fails
4	N6 share below 0.50	0.015	Point criterion met
5	Underprediction in at least four bands	Intervals exclude 1 in one band	Fails interval reading, meets point reading
6	Transport within 20 percent, independence error above 20 percent	+66%, -28.9%, three units	Excess-scale criterion fails, raw scale differs
7	External estimate and lower limit above 1	1.705 for likelihood	Required interval unavailable
8	At least eight ratios in [1.4, 3.0]	Ten, range [1.571, 1.732]	Lower threshold can't discriminate
9	Cross-format diagnostic above 1.2, lower limit above 1	0.921 and 0.965	Neither criterion met
10	Frobenius distance above 0.5	5.59 and 6.49	Criterion met with fill convention
11	More than ten percent of pairs with predicted flip probability above 0.05	455 of 1,500, or 30.3 percent	Observed-gap calculation meets criterion

depends on a post hoc choice to compare excess inflation. We retain the numerical results while distinguishing these qualifications from tests of verified prospective hypotheses.

Variation across model sizes Each size band contains 25 configurations, one per recipe. Its clusters are therefore singletons, without within-band replication of sizes inside recipes. Only the 530M interval excludes one in either phenotype, and the intervals don't support a ranking of model sizes. All five margin point estimates and four accuracy point estimates exceed one, so prediction 5 passes a point-estimate interpretation and fails the stricter interval interpretation used in the original scorecard.

Table 8. Inflation estimates by model size across the 25 recipes. The intervals use the wild cluster bootstrap-t calculation described in Appendix D.

Size	Λ_M	95% interval	Λ_A	95% interval
150M	1.101	[0.781, 1.344]	1.028	[0.878, 1.165]
300M	1.350	[0.638, 1.803]	1.080	[0.836, 1.288]
530M	1.514	[1.064, 1.837]	1.219	[1.083, 1.338]
750M	1.110	[0.686, 1.407]	0.987	[0.310, 1.320]
1B	1.149	[n/a, 1.751]	1.205	[0.805, 1.497]
Pooled, $N = 125$	1.244	[1.143, 1.338]	1.078	[0.993, 1.157]

The lower squared-ratio bound for 1B margins falls below zero, so its square-root endpoint is unavailable. Figure 2 reports N1 calibration at six effect sizes with 2,000 replicates each, using the 2.5th and 97.5th percentiles of estimates as dispersion bands. Those bands describe simulated estimator dispersion rather than confidence-interval coverage.

Permutation comparison N2 was proposed as an estimator-bias check rather than the primary inferential procedure. In the reported 500 permutations, none reaches the observed margin estimate, and 0.028 reach the accuracy estimate. This reassignment holds the observed configurations fixed while independently permuting trait-specific run labels. The primary bootstrap instead targets

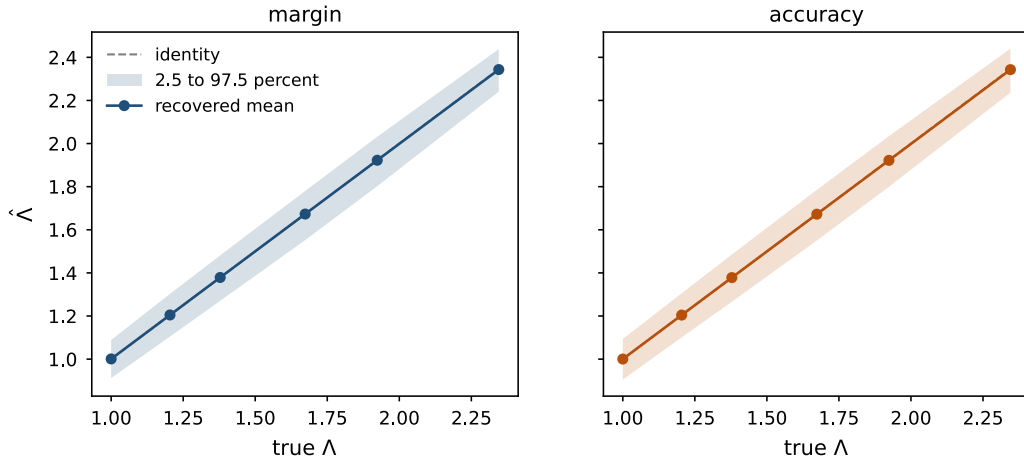


Figure 2. Reported simulation recovery across six specified covariance settings for both score scales. The shaded bands describe replicate dispersion and don’t establish interval coverage under other data-generating assumptions.

variation across recipe clusters, so the two procedures use different randomisation and sampling assumptions. The accuracy permutation result and the interval containing one should therefore remain separate reported findings.

Associations with scoring format The cross-format diagnostic retains the diagonal sum and only those off-diagonal terms whose traits have different formats. It has a null reference of one when those retained covariances vanish, but the masked sum needn’t represent the variance of a realisable benchmark average. Because each benchmark has only one format, the comparison can’t identify a causal effect of format.

The separate calculation reproduces the full estimate to four decimal places before retaining 54 of the 90 ordered off-diagonal pairs. It gives 0.921 with interval 0.801 to 1.035 for margins, and 0.965 with interval 0.848 to 1.063 for accuracy. Both fall below the proposed 1.2 threshold, and the margin result falls below the 1.153 expected from uniformly distributed covariance contributions under the comparison used here.

BoolQ is the only yes/no benchmark, so all its off-diagonal pairs cross a format boundary. Its margin seed variance is 1.43×10^{-3} , compared with 4.34×10^{-4} for the next-largest trait. Excluding BoolQ gives a cross-format margin diagnostic of 1.006 with interval 0.972 to 1.039, compared with an unrestricted nine-trait estimate of 1.786 with interval 1.689 to 1.877.

Among the nine traits with defined margin correlations, excluding WinoGrande, the mean correlation is 0.624 across sixteen within-format pairs and 0.007 across twenty cross-format pairs. The three cloze traits give margin inflation of 1.225 with interval 1.064 to 1.367. Excluding ARC-Easy and ARC-Challenge from the multiple-choice block leaves inflation of 1.479 with interval 1.416 to 1.541. These associations show concentration within the larger format groups, while benchmark content and format remain confounded. A rank-one model with trait-specific loadings can also produce uneven contributions across those groups.

Table 9 applies $\bar{r}_E = 0.0181$ while holding trait standard deviations fixed. Its adjusted column applies to the weighting of this battery, with flip probabilities interpreted under the stated Gaussian comparison model. The held-out accuracy point errors favour the full matrix over independence, 0.0105 versus 0.0143, without an interval for their difference. Every predictor also receives held-out seed standard deviations unavailable without replication, so these results don’t establish an operational replacement for replicated runs. The three-unit PolyPythias accuracy estimate of 2.286 differs from 1.078 and lacks an interval, which further limits transport of this calculation.

Table 9. Accuracy-point calculations at the median 300M configuration. The four- and seven-trait rows evaluate the same ten-trait quantities at different K values and aren't additional measured batteries.

K	Independent σ_{agg}	Adjusted σ_{agg}	K_{eff}	P_{flip} , one-point gap
4	0.754	0.774	3.79	0.181
7	0.570	0.600	6.31	0.119
10	0.477	0.514	8.60	0.085

Table 10. Reported null and sensitivity results with replicate standard deviations. N3 and N4 use the observed correlation and therefore describe sensitivity rather than a zero-correlation null.

Test	Structure	$\hat{\Lambda}_M$	SD	$\hat{\Lambda}_A$	SD
N1	Zero correlation with observed counts and reliabilities	1.0009	0.046	1.0004	0.048
N2	Independent trait-specific run-label permutations	1.0010	0.048	0.9984	0.039
N1	$\bar{r}_E = 0.05$, true $\Lambda = 1.2042$	1.2048	0.051	1.2044	0.053
N1	$\bar{r}_E = 0.10$, true $\Lambda = 1.3784$	1.3788	0.055	1.3782	0.057
N1	$\bar{r}_E = 0.20$, true $\Lambda = 1.6733$	1.6730	0.059	1.6724	0.061
N1	$\bar{r}_E = 0.30$, true $\Lambda = 1.9235$	1.9226	0.059	1.9220	0.061
N1	$\bar{r}_E = 0.50$, true $\Lambda = 2.3452$	2.3433	0.052	2.3429	0.053
N3	Fifty independent item re-splits	1.2459	0.002	1.0843	0.008
N4	Zero mediation with pooled-slope regression	1.0910	0.096	1.1045	0.062
N5	Zero correlation plus gain, $\sigma_g = 0.05$	1.1477	0.050	0.9986	0.048
N6	Zero correlation plus default-versus-auxiliary offset	1.0038	0.044	1.0030	0.047
N6	Same offset with auxiliary-run contrast	1.0004	0.062	1.0003	0.065

Performance correlation as a noise proxy The reported Frobenius distances between R_E and R_P are 5.59 for margins and 6.49 for accuracy. Negative estimated variances leave eighteen margin and 34 accuracy ordered off-diagonal correlations undefined. The implementation fills these entries with zero before differencing, including undefined diagonal entries compared with unit diagonals in R_P . Those diagonal substitutions contribute 1.0 and 2.0 to the squared distances.

Repairing only those diagonals gives distances of 5.499 and 6.339. Restricting the comparison to jointly defined entries gives 4.374 and 4.593, which still exceed the planned threshold of 0.5. The threshold alone offers limited discrimination between different ten-trait matrices, so the entrywise comparisons provide a more useful description.

The off-diagonal association between the matrices is 0.756 for margins and 0.345 for accuracy. Mean defined R_E entries are 0.281 and 0.273, compared with all-pair R_P means of 0.750 and 0.754. On matching pairs, the R_P means are 0.743 and 0.769. The all-pair comparison therefore differs in scale by factors of approximately 2.7 and 2.8. The unadjusted proxy predicts a mean margin aggregate standard deviation of 0.0072, compared with the reported 0.0057.

An additional exploratory baseline, P2 R, rescales R_P off-diagonals by the training-fold ratio of mean noise correlation to mean performance correlation. Its mean factors are 0.366 for margins and 0.370 for accuracy, compared with whole-population ratios of 0.375 and 0.362. On the same five recipe folds, its squared prediction errors are 0.0131 and 0.0342. It improves on P2 but remains 2.3 times worse than P1 on margins and worse than independence on accuracy. The direction relative to independence repeats across three fold seeds, although five folds don't establish uncertainty for that ranking. This rescaling also requires replicated runs to estimate its factor.

Recipe, size, and trait sensitivity Removing each recipe in turn gives margin estimates from 1.207 to 1.256 and accuracy estimates from 1.065 to 1.099. The maximum changes from the complete battery are 0.037 and 0.021. Removing each size gives ranges of 1.178 to 1.289 and 1.047 to 1.115, compared with individual-band ranges of 1.101 to 1.514 and 0.987 to 1.219. These overlapping intervals don't establish a size trend in the effect.

Trait removal changes the estimates more than recipe or size removal. Excluding BoolQ gives 1.786 with interval 1.689 to 1.877 for margins, and 1.558 with interval 1.412 to 1.687 for accuracy. Excluding CommonsenseQA gives the smallest point estimates, 1.096 and 1.014. BoolQ contributes 68 percent of the estimated margin variance trace and 84 percent of the accuracy trace, while its average covariance with other traits is near zero. It therefore contributes mainly to the denominator of the excess ratio.

Table 11. Trait-removal estimates across all 125 configurations in both phenotypes. Each removal row uses the nine remaining traits in the battery.

Removed trait	Λ_M	Λ_A
BoolQ	1.786	1.558
WinoGrande	1.246	1.102
HellaSwag	1.244	1.076
PIQA	1.243	1.073
MMLU	1.205	1.068
ARC-Challenge	1.195	1.039
OpenBookQA	1.178	1.074
ARC-Easy	1.161	1.046
Social IQa	1.139	1.032
CommonsenseQA	1.096	1.014
None	1.244	1.078

The variance-share participation ratios are 1.97 for margins and 1.42 for accuracy on traits with defined seed variance. These quantities measure concentration of variance across traits, while the noise-correlation participation ratio of 3.57 measures concentration across matrix directions. They describe different aspects of the same battery’s weighting.

Standardising traits by estimated seed standard deviation gives 1.803 on nine margin traits and 1.707 on eight accuracy traits. These point diagnostics omit traits with nonpositive variance estimates and divide by uncertain standard deviations, with seven accuracy traits at or below item noise. The analysis supplies no resampling distribution for these quantities, so we don’t attach confidence intervals or score them against the planning thresholds.

Removing defined traits from the standardised batteries gives margin estimates from 1.625 to 1.910 and accuracy estimates from 1.388 to 1.780. BoolQ removal gives 1.910 and 1.761 under this standardisation. Selecting traits by positive estimated variance can change the target, and it could raise the mean correlation if excluded low-signal traits also have low true correlations. We haven’t tested that additional assumption, and subset stability doesn’t resolve uncertain denominators.

The raw trait-removal spans are 0.690 for margins and 0.544 for accuracy, compared with interval widths of 0.195 and 0.164. The resulting ratios of 3.5 and 3.3 compare different measures of spread and aren’t uncertainty decompositions. Applying the delete-one-trait jackknife formula produces scales of 0.554 and 0.456, compared with sampling standard errors of 0.045 and 0.038. Their ratios are 12.3 and 12.1, partly because the jackknife multiplier is 2.85. Comparing the unscaled dispersion instead gives ratios of 4.33 and 4.25.

Omitting the BoolQ removal value from that calculation gives scales of 0.138 and 0.073, or 3.07 and 1.93 times the reported sampling standard errors. These are descriptive composition diagnostics, because the ten fixed benchmarks aren’t established exchangeable draws from a benchmark population. Adding the original jackknife scales and sampling standard errors in quadrature gives 0.556 and 0.457. A normal calculation then gives ranges of 0.15 to 2.33 and 0.18 to 1.974, both including one. Repeating that arithmetic without the BoolQ removal value gives 0.959 to 1.529 and 0.917 to 1.239.

Those calculations aren’t validated confidence intervals for a benchmark population. They can double-count sampling variation, and their normal approximation behaves poorly near the parameter boundary. For example, $\Lambda = 0.15$ corresponds to $\bar{r}_E = -0.109$, only approximately 0.003 above $-1/9$. We retain the fixed-battery target and report composition sensitivity without treating these scales as an identified decomposition of total uncertainty.

Changing trait count also changes Λ at fixed effective covariance coefficient. At $\bar{r}_E = 0.061$, the nine-trait reference is 1.219 rather than the ten-trait 1.244, while the corresponding accuracy values are 1.070 and 1.078. Four margin removals and five accuracy removals exceed those mechanical references. On eight traits, standardised accuracy inflation of 1.707 implies 2.75 effective benchmarks, while the raw accuracy effective coefficient of 0.0181 implies 7.10 effective benchmarks. Comparing 2.75 with 7.10 holds trait count fixed, unlike comparison with the ten-trait value of 8.60.

The ARC pair has an estimated margin correlation of 0.801. Removing both benchmarks gives margin inflation of 1.118 with interval 1.059 to 1.174, and accuracy inflation of 1.012 with interval 0.929 to 1.086. The pair affects the estimate, although the remaining margin interval still excludes one.

Interval construction and cluster sensitivity A percentile bootstrap over 125 configurations gives intervals of 1.069 to 1.426 for margins and 0.976 to 1.182 for accuracy. Both preserve the qualitative finding of margin separation and accuracy non-separation from one. Comparing squared standard errors, using 0.045 versus 0.093 and 0.038 versus 0.052, gives ratios of 0.24 and 0.53. The implementation separately reports numerator-based design effects of 2.06 and 1.26 from $1 + 4\rho_{ICC}$, compared with the planned 1.60.

These ratios describe different quantities and aren't contradictory estimates of one design effect. The numerator calculation uses T_c , while linearisation uses $T_c - \hat{\Lambda}^2 U_c$. The interval procedures also change resampling unit, studentisation, and denominator treatment together. We consequently don't infer a single intraclass correlation from their relative widths. The bootstrap-t addresses concerns associated with few-cluster approximations, as discussed by Bell and McCaffrey (2002), but it doesn't guarantee coverage at 25 clusters.

Observed gaps and predicted rank reversals Using accuracy noise $\sigma_{agg} = 0.00514$, the Gaussian calculation gives $P_{flip} > 0.05$ for a true gap below 0.01196. Among the $5\binom{25}{2} = 1,500$ observed same-size pairs, 455 have observed gaps below this threshold, or 30.3 percent. The proportions range from 28.7 to 33.3 percent across bands. Substituting observed gaps meets the stated numerical criterion, but noisy observed gaps can move pairs across the threshold in either direction. This count therefore isn't a lower bound on the fraction of true gaps below the threshold, and it doesn't directly measure realised reversal frequency.

Adjacent-checkpoint variation We compare all nine 160M PolyPythias runs at steps 143000 and 142000 on a nested 1,800-item sample. The check includes nine non-MMLU traits and fixes items, scoring code, and batching. MMLU's 57 subjects would increase the forward passes approximately threefold to include one additional trait, so this check excludes it.

The margin battery's between-run standard deviation is 0.00513, while the root-mean-square checkpoint shift is 0.00456, a ratio of 0.890. Accuracy gives 0.00451 and 0.00710, a ratio of 1.573. Across the nine runs, final and preceding checkpoint scores correlate 0.573 for margins and -0.070 for accuracy. These results show substantial checkpoint sensitivity in this restricted sample, without separating stochastic trajectory variation from systematic progress across 1,000 steps.

The estimator describes run variation at the scored checkpoint, which can include trajectory effects correlated across traits. It doesn't isolate initialisation from these effects in the available design. This comparison uses 1,800 rather than 4,755 items, nine runs, and a final checkpoint, whereas DataDecide uses common steps during training. We haven't measured the corresponding checkpoint sensitivity in DataDecide. BoolQ has the largest trait-level values in this check, with between-run standard deviation 0.0438 and root-mean-square shift 0.0370.

Simulation and permutation calibration The linearised margin standard error of 0.045 is close to the N1 null dispersion of 0.046. Accuracy gives 0.038 against the simulated 0.048, a difference of approximately 21 percent. The reported interval half-widths are 0.0974 and 0.082, approximately 2.2 standard errors compared with the $t(24)$ critical value of 2.06. Substituting the accuracy simulation dispersion gives a descriptive interval of 0.979 to 1.178, which also includes one. This substitution isn't a calibrated alternative interval because the simulated and observed populations differ.

A separate implementation of the interval definitions uses 25 clusters of five configurations, three runs, and ten traits. It assumes exchangeable seed covariance, independent item noise across halves, and recipe heterogeneity in seed variance. Across sixteen cells formed from four true correlations between zero and 0.10 and four noise settings, 200 datasets per cell give empirical coverage from 0.920 to 0.975. Each cell lies within two Monte Carlo standard errors of 0.95, and estimated inflation is within 0.01 of truth in fifteen cells. Four cells with dispersion 0.044, compared with observed null dispersions 0.046 and 0.048, have coverage 0.950, 0.935, 0.975, and 0.960. Their interval widths range from 0.203 to 0.238, compared with the reported margin width of 0.195. This limited simulation checks the procedure under its specified populations and doesn't validate all departures present in the study.

The analysis also reports 1,000 trait-specific run-label permutations, each using 4,999 bootstrap draws. The resulting intervals contain the reference value one in 0.925 of margin permutations and 0.954 of accuracy permutations. Margin intervals fall below one in 0.049 and above one in 0.026 of permutations, compared with nominal tail probabilities of 0.025. The lower-tail imbalance matters for interpreting an upper bound below the proposed threshold of 1.349, where the reported upper endpoint leaves only 0.011 of separation, approximately one quarter of a margin standard error.

These are conditional randomisation calibration results, contingent on the exchangeability assumed by the permutations. They aren't coverage measurements for the unknown nonzero covariance of the original population. The cluster-t comparator has reported coverage of 0.881 and 0.941, with above-one rates of 0.044 and 0.023. In the same permutations, no margin estimate reaches the observation, while the accuracy upper-tail fraction is 0.021, close to 0.028 in the earlier 500 draws. With a plus-one correction, zero exceedances among 1,000 draws give a Monte Carlo value of 1/1001, rather than zero.

The reported delete-one-recipe jackknife standard errors are 0.049 and 0.039. Applying a two-sided 95 percent $t(24)$ critical value to those rounded inputs gives approximate intervals of 1.143 to 1.345 for margins and 0.998 to 1.158 for accuracy. The earlier narrower intervals use a multiplier close to a 90 percent interval, which prevents their use as 95 percent comparisons. The corrected accuracy interval includes one, consistent with the primary interval.

Across the 1,000 permutations, estimator standard deviations are 0.0488 for margins and 0.0382 for accuracy. The earlier 500-draw values are 0.048 and 0.039, compared with N1 values of 0.046 and 0.048. The accuracy discrepancy therefore concerns simulated versus conditional permutation populations, rather than establishing an error in the observed bootstrap scale.

Availability of a separate competence measurement The desired gain control requires per-run loss on held-out text separate from the benchmark items. The DataDecide fields inspected in the source analysis don't supply it. The PolyPythias evaluation files considered there cover three sizes at early and final checkpoints, but contain accuracies, likelihood differences, and LAMBADA perplexity on the same benchmark items. That perplexity doesn't provide the required independent measurement scale.

The Mistral release of Karamcheti et al. (2021) provides five GPT-2 Small and five GPT-2 Medium runs with 610 checkpoints each, including WikiText-103 and OpenWebText validation perplexity. The source search didn't find the corresponding per-run benchmark battery. This search limitation leaves the gain interpretation unresolved and doesn't turn simulator shares into causal bounds.

Transport to PolyPythias The planned transport analysis scores nine runs per configuration on a nested 4,755-item battery. For three sizes, the source pipeline gives margin inflation of 1.40574 compared with 1.24395 for DataDecide, an excess ratio of 1.66323. A scoring-kernel rewrite retains the margin estimate to five decimals, with the preceding excess ratio reported as 1.66321. The configuration table requires the second-arm output rather than inspection of the DataDecide reduction alone.

On excess inflation, the DataDecide prediction falls approximately 40 percent below PolyPythias, while the raw inflation difference is approximately twelve percent. The planning document names aggregate standard deviation without resolving this scale choice. The excess-based failure assessment is therefore post hoc, and the raw inflation comparison falls inside the proposed twenty-percent band.

A later exploratory extension includes the 14M and 31M configurations, each with nine runs at step 143000 on the same 4,755 items. Within that rescore, margin inflation changes from 1.41372 at three sizes to 1.26199 at five sizes, while the excess ratio changes from 1.69592 to 1.07395. Accuracy changes from 2.28589 to 1.71085, compared with DataDecide’s 1.07837. Only the three-size accuracy point estimate exceeds the stated detection threshold of approximately 2.05, and the analysis supplies no interval for either accuracy estimate.

The margin configuration-bootstrap intervals are 0.939 to 1.781 at three sizes and 0.927 to 1.537 at five sizes. Both contain one and DataDecide’s 1.24395, while the standard error falls from 0.190 to 0.150. These intervals limit conclusions about transport under either point-estimate assessment. The rescore used one T4 with a 6,000-token cap instead of 30,000 tokens, and its three-size margin value of 1.41372 differs from the pipeline’s 1.40574 by 0.6 percent. The excess-ratio difference is approximately two percent, with hardware and batching changed together. The three-versus-five comparison holds those conditions fixed within one job, and the source supplement describes 45 per-run score files from that job.

Table 12. Five-fold prediction error for log aggregate seed standard deviation. Every model receives held-out trait standard deviations, so the comparison measures correlation prediction with variance estimation fixed.

Model	Parameters	Margin error	Accuracy error
P0 independence	K	0.0358	0.0143
P1 rank-one competence	$K + 1$	0.0057	0.0140
P1 G rank-one plus gain	$2K + 2$	0.0063	0.0260
P2 performance correlation	K	0.1073	0.1682
P3 full covariance	$K(K + 1)/2$	0.0068	0.0105

Table 12 gives fold-mean point errors without uncertainty intervals for their differences. P1 and P1 G have lower point errors than P3 on margins and higher errors on accuracy, while P0 and P2 have higher errors on both scales. P2 uses performance correlation from the held-out fold itself because its calculation needs no seed replication. Negative held-out variance estimates are clipped to zero before model selection, which also applies to independence. On the complete population this clips one margin and two accuracy traits, increasing the aggregate covariance sum by 0.3 and 4.8 percent, while the scored target retains the original estimates. Shared clipping doesn’t guarantee that it leaves model rankings unchanged.

External random-seed panel The signal-and-noise runs don’t share models with the DataDecide analysis, although the releases come from one laboratory, use a shared scoring harness, and overlap in all ten study traits. The check averages estimates over twenty released checkpoints and uses ten initialisation runs or nine data-order runs. It supplies single-unit estimates without cross-half correction or uncertainty intervals, which limits comparison with the corrected primary estimator.

Table 13. Inflation estimates in the external random-seed panel across twenty checkpoints. The leave-one-out adjustment controls each trait deviation with the mean deviation of the other traits.

Seed variation	Phenotype	K	Λ	Adjusted Λ	$\bar{r}_E$
Initialisation	bits per byte	18	1.705	1.464	0.112
Initialisation	accuracy per character	8	0.989	1.105	-0.003
Data order	bits per byte	18	1.625	1.763	0.096
Data order	accuracy per character	8	0.868	1.059	-0.035

The likelihood estimates exceed one under both seed mechanisms, while the unadjusted accuracy values lie near or below one. Adjustment increases three of the four estimates and reduces only initialisation likelihood, from 1.705 to 1.464. That pattern doesn’t support a stable interpretation of the adjustment at one unit. Independent item noise would attenuate covariance ratios toward one under the relevant assumptions, but this arm doesn’t measure the amount of attenuation or

verify those assumptions. The external comparison remains suggestive rather than independent confirmation of the primary magnitude.

C.1 GAIN CALIBRATION AND RECIPE SPLIT SENSITIVITY

The default gain simulator applies a shared margin scale across traits. The matched simulator instead samples each trait’s margins from its empirical distribution. At $\sigma_g = 0.05$, the two reported excess shares differ by approximately a factor of seven. Subtracting estimated leakage variance 1.91×10^{-3} from total gain variance 3.67×10^{-3} gives a gain standard deviation of approximately 0.042. This subtraction depends on the variance decomposition assumed for the covariate.

Table 14. Gain-simulator sensitivity relative to the observed margin excess of 0.244. These shares compare specified simulations and don’t bound a causal contribution in the observed runs.

σ_g	Default $\hat{\Lambda}_M$	Excess share	Matched $\hat{\Lambda}_M$	Excess share
0.01	1.007	0.029	1.001	0.003
0.02	1.026	0.108	1.003	0.014
0.03	1.058	0.236	1.008	0.031
0.042, estimated	1.108	0.443	1.014	0.059
0.05, default	1.149	0.609	1.020	0.081
0.10	1.465	1.905	1.062	0.256

The default 0.05 cell differs by approximately 0.0009 from the corresponding Table 10 result before rounding, less than half the reported Monte Carlo standard error for 500 replicates. The G5 assessment uses that default specification, which fails its threshold. Accuracy estimates stay within 0.004 of one in these simulations, but positive multiplication preserves signs by construction. Their proximity to one therefore doesn’t establish absence of a gain mechanism for observed accuracy variation.

For a recipe subset S , the estimator is $\hat{\Lambda}_S = \sqrt{\sum_{c \in S} T_c / \sum_{c \in S} U_c}$, because each configuration’s components require no other configurations. The analysis enumerates all $\binom{25}{17} = 1,081,575$ possible seventeen-recipe estimation sets. It reports completing this sweep in thirteen seconds on one CPU core. The reconstructed split uses the plan’s master seed under the released implementation, but the plan didn’t specify a complete reconstruction rule and no contemporaneous split artifact was recovered.

Table 15. Exhaustive sensitivity to the choice of seventeen estimation recipes. Every admissible subset gives a finite point estimate under both phenotypes.

Quantity	Margin	Accuracy
All 25 recipes	1.244	1.078
Reconstructed planned partition	1.199	1.099
Minimum	1.037	0.963
5th percentile	1.176	1.035
Median	1.246	1.078
95th percentile	1.285	1.124
Maximum	1.317	1.191
Effective-count range	[5.77, 9.31]	[7.05, 10.77]
Planned threshold	1.349	1.40
Subsets reaching threshold	0	0
Subsets below one	0	1,393, or 0.13%

No subset reaches either proposed point threshold in this enumeration. The sweep addresses sensitivity to recipe selection for the reported estimator, while it can’t restore the absent prospective screening separation or remove other analysis choices made with the data visible.

D IMPLEMENTATION DETAILS

D.1 DATA STREAMING

The implementation streams 122.9 GB of gzipped JSONL tarballs from the Hugging Face release. It retains the item identifier, correct-answer margin, and accuracy bit for the ten traits at the common checkpoint. Each run contains 37,682 items and approximately 30 MB of compressed output. Reduction of 375 runs produces approximately 48 MB of NumPy archives, with one array per run.

A frozen master seed generates a balanced permutation of item order within each trait. The split uses that permutation rather than a hash of item identifiers, and all runs within an arm reuse the same trait-specific assignment. The PolyPythias subsample therefore has its own split, rather than an item-hash split shared across arms. Regenerating the split requires the recorded item ordering and randomisation specification, without downloading the full tarballs.

D.2 INTERVAL CALCULATION

We state the source calculation in terms of the squared ratio,

$$\hat{\theta} = \frac{\sum_c T_c}{\sum_c U_c}, \quad \hat{\Lambda} = \sqrt{\hat{\theta}}.$$

For each recipe g among $G = 25$, the linearised contribution and standard error are

$$\psi_g = \frac{\sum_{c \in g} (T_c - \hat{\theta} U_c)}{\sum_c U_c}, \quad \widehat{\text{SE}} = \left(\sum_{g=1}^G \psi_g^2 \right)^{1/2}. \quad (20)$$

This standard error belongs to the squared-ratio scale used for studentisation. A reported standard error on the Λ scale instead requires the corresponding delta-method conversion.

For draw b , independent recipe signs satisfy $v_{bg} \in \{-1, +1\}$ with equal probabilities. Every configuration takes the sign assigned to its recipe, and the pseudo-numerator becomes

$$T_c^* = \hat{\theta} U_c + v_{bg(c)} (T_c - \hat{\theta} U_c).$$

The denominator remains fixed at the observed $\sum_c U_c$. We recompute $\hat{\theta}_b^* = \sum_c T_c^* / \sum_c U_c$ and the bootstrap standard error from the residuals $T_c^* - \hat{\theta}_b^* U_c$. The bootstrap statistic is $t_b^* = (\hat{\theta}_b^* - \hat{\theta}) / \widehat{\text{SE}}_b^*$, and the source calculation uses 4,999 draws with bootstrap seed zero.

Writing q_α for the empirical quantile of these statistics, the interval on θ is

$$\left[\hat{\theta} - q_{0.975} \widehat{\text{SE}}, \hat{\theta} - q_{0.025} \widehat{\text{SE}} \right].$$

Square-root transformation gives endpoints on the inflation scale when both endpoints are nonnegative. Monotonicity preserves the ordering of available endpoints, but it doesn't establish the original interval's coverage.

A draw with zero bootstrap standard error has no finite studentised statistic and is discarded. The routine requires at least $\max(100, 0.9B)$ surviving draws before reporting an interval. A negative lower squared-ratio endpoint is recorded as missing rather than truncated, which explains the absent 1B margin lower endpoint in Table 8. The configuration percentile bootstrap instead resamples all 125 configurations with replacement and recomputes both sums, while ignoring recipe membership.

The planning document specified wild cluster bootstrap-t inference with $t(16)$ critical values for seventeen estimation recipes. The realised analysis uses empirical bootstrap quantiles over all 25 recipes, following the failure to implement the planned split. The analysis record states that a $t(G-1)$ variant accompanies the bootstrap implementation. These alternatives differ in more than a change of cluster count.

D.3 PRACTICAL CONSIDERATIONS

Within-configuration centring removes the shared performance mean before forming cross-half products. With only three runs, raw products would include a large contribution from mean competence rather than measuring covariance of deviations. The contrast representation in Appendix D.7 enforces that centring at the estimator input.

The permutation procedure reassigns run labels independently within each trait. Applying one joint permutation to all traits preserves cross-trait pairing and therefore doesn't generate this independence null. Identity permutations remain valid members of the randomisation distribution and shouldn't be excluded. For three runs, independent identities across all K traits have probability 6^{-K} within one configuration.

Gzipped tarball members require sequential reading from their starts. The reduction therefore extracts the required fields on one pass, since an omitted field requires another complete stream. The default-versus-auxiliary step mismatch at 530M and 750M motivates the auxiliary-two-versus-auxiliary-three contrast, with checkpoint agreement checked separately for both auxiliary runs.

The pipeline lists the 375 checkpoint directories once, applies exponential backoff to rate limits, and caches the listing. Before estimation, it regenerates the common-step table from that cache and compares it with the initial version. This check can detect changes in the recorded step selection, but a cached listing alone can't detect new remote changes without a refreshed comparison.

D.4 COMPUTE

Table 16. Compute budget reported in the planning document for both analysis arms. These figures describe the plan rather than measured costs for every listed operation.

Operation	Budget	Scope
Arm 2 held-out bits per byte	6.0 GPU hours	27 runs, 4,755 items
Calibration simulations N1, N5, N6	1.0 GPU hours	2,000 replicates, six effects
160M checkpoint check	0.5 GPU hours	Adjacent checkpoints
Contingency	2.5 GPU hours	Reserved capacity
Total GPU	10.0 GPU hours	Planned total
Streaming and reduction	18 CPU hours	122.9 GB, one pass
Estimation, bootstrap, permutation	7 CPU hours	Includes 200 planned splits

The primary run used one Kaggle CPU core for Arm 1, reporting 6.0 hours for estimation and null checks. Four sensitivity tasks took another 1.6 hours, including gain calibration and fifty re-splits rather than the 200 budgeted. The exact recipe sweep took thirteen seconds, and the later 160M checkpoint check took 423 seconds of scoring on one T4. Arm 1 performs no forward passes and reports no GPU use.

The transfer procedure streams 25 monolithic tarballs totalling 122.9 GB while keeping peak disk use below 10 GB. Arm 2 performs teacher-forced fp16 scoring with zero-shot contexts and no exemplars. Most items use one context, while WinoGrande uses one per choice. Reported context lengths include 199 tokens for ARC-Challenge and 937 for BoolQ.

The budget assumed two 16 GB cards, but the reported scoring jobs instead received one 46 GB L40S each on a Slurm cluster. Each of 27 jobs finished in under two minutes, with aggregate reported use below one card-hour rather than the budgeted six. Gold-token scoring materialises a float32 log-softmax tensor over a 50,304-token vocabulary. A 30,000-token batch requires approximately 6 GB for that tensor alone.

A configuration rescore fit on a 23 GB card at the planned token budget, while the 16 GB T4 checkpoint check used 6,000 tokens. Changing the budget changes batch grouping, which can affect floating-point results even when the mathematical scoring rule stays fixed. The transport rescore changed hardware and token budget together and shifted inflation by 0.6 percent. CPU handles the reduction and statistical analysis, with model scoring and calibration kernels as the reported exceptions.

D.5 RELEASE PROVENANCE

The null checks examine mechanisms that can affect covariance estimates, but they don't independently establish the provenance of the three released runs. Differences in tokeniser builds, harness revisions, or other training conditions could contribute to the default-versus-auxiliary contrast if

those conditions weren't held fixed. The auxiliary-run contrast also relies on documentation that the relevant settings match between its two runs.

The step table and disagreement checks expose some inconsistencies without replacing a release-side audit. Attribution of the observed contrasts to the documented seed changes therefore depends on the release documentation. This limitation applies independently of the estimator's behaviour on simulated or permuted data.

D.6 SENSITIVITY TO GPU ARCHITECTURE

The primary Arm 2 scores use one card type. A separate Turing-versus-Ada comparison on one configuration reports mostly identical item scores, a mean absolute margin difference of 1.0×10^{-3} , and a largest difference of 3.5×10^{-2} . Relative to a margin standard deviation of 0.391, the mean difference has ratio 2.6×10^{-3} .

Eighteen of 4,755 accuracy bits change, or 0.38 percent. Their median absolute margin is 1.3×10^{-3} , and battery accuracy changes from 0.430074 to 0.430494, a difference of 4.2×10^{-4} . That difference is approximately eight percent of the reported accuracy seed standard deviation. This single comparison demonstrates sensitivity at that scale, but it doesn't establish a universal portability threshold for fp16 kernels.

Arm 1 uses released per-item outputs without recomputing forward passes, so this study doesn't add the same hardware variation to those scores. Shared hardware across Arm 2's 27 runs reduces one source of variation but doesn't independently validate accuracy across architectures.

D.7 CENTRING THROUGH RUN CONTRASTS

Raw cross-half products contain squared configuration means, which the analysis record reports can inflate the ratio by roughly three orders of magnitude. The described implementation projects scores onto an orthonormal basis of mean-zero run contrasts. Each basis row annihilates the all-ones vector, making raw and centred inputs equivalent under projection. Runtime checks test row sums and invariance of $\sum_c T_c$ under a shared configuration shift on both halves, addressing centring errors without testing measurement or exchangeability assumptions.